

CHATHAM ISLAND, New Zealand

WMO: 939850

Lat: 43.8167S

Lon: 176.4750W

Elev: 12

StdP: 101.18

Time Zone: 12.75 (NZC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.0	3.9	-0.7	3.6	4.8	0.3	3.9	5.3	18.2	8.8	16.4	8.1	4.0	180	0.750

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.4	20.8	17.6	19.8	17.1	18.9	16.5	18.4	19.9	17.7	19.0	17.2	18.3	7.7	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	12.7	19.1	17.2	12.3	18.4	16.7	11.9	17.9	52.2	20.0	50.2	19.2	48.4	18.4	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
15.7	14.1	12.6	1.1	21.9	1.2	1.4	0.2	22.9	-0.5	23.7	-1.2	24.5	-2.1	25.5	
			0.2	19.5	1.0	0.7	-0.6	20.0	-1.2	20.4	-1.7	20.8	-2.5	21.3	
			DB												
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.6	14.8	15.3	14.5	12.6	10.9	9.2	8.3	8.7	9.5	10.5	11.6	13.7	(6)
(7)		DBStd	3.20	2.15	2.10	2.15	2.27	2.05	2.14	2.01	1.90	2.03	2.01	2.23	2.38	(7)
(8)		HDD10.0	229	0	0	0	3	13	41	62	48	33	19	9	1	(8)
(9)		HDD18.3	2459	110	88	120	172	230	274	312	297	265	244	203	144	(9)
(10)		CDD10.0	817	149	148	139	82	42	17	8	9	18	33	56	116	(10)
(11)		CDD18.3	4	1	2	1	0	0	0	0	0	0	0	0	1	(11)
(12)		CDH23.3	1	1	1	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	6.6	6.4	6.0	6.3	6.5	6.3	7.0	6.5	6.7	7.1	7.1	7.0	6.6	(14)
(15)	Precipitation	PrecAvg	896	58	62	79	81	95	105	77	80	67	64	57	64	(15)
(16)		PrecMax	1075	169	121	178	152	181	164	151	155	125	122	191	165	(16)
(17)		PrecMin	654	10	11	14	32	10	66	40	18	9	23	14	17	(17)
(18)		PrecStd	118	37	31	41	34	35	26	29	35	31	27	38	38	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.9	22.4	21.1	18.8	16.6	14.8	13.9	14.3	15.3	17.1	19.3	21.4	(19)
(20)			MCWB	17.8	18.7	17.9	16.8	15.0	13.6	12.5	12.3	12.9	13.2	15.3	17.3	(20)
(21)		2%	DB	20.5	21.0	19.6	17.7	15.7	13.9	13.1	13.0	14.1	15.5	17.4	19.6	(21)
(22)			MCWB	17.3	17.9	17.2	16.0	14.6	12.9	12.0	11.8	12.3	13.0	14.3	16.7	(22)
(23)		5%	DB	19.4	19.7	18.6	16.8	15.1	13.3	12.3	12.3	13.3	14.6	16.2	18.3	(23)
(24)			MCWB	16.8	17.2	16.5	15.4	14.1	12.3	11.3	11.2	12.0	12.6	13.7	15.9	(24)
(25)		10%	DB	18.4	18.7	17.8	16.1	14.4	12.6	11.6	11.7	12.7	13.7	15.2	17.2	(25)
(26)			MCWB	16.1	16.7	16.1	14.9	13.4	11.7	10.7	10.7	11.6	12.1	13.3	15.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.0	19.6	18.7	17.4	15.8	14.1	13.1	12.9	13.6	14.4	16.1	18.2	(27)
(28)			MCDWB	20.9	21.3	20.1	18.3	16.2	14.4	13.5	13.4	14.4	15.6	18.6	19.9	(28)
(29)		2%	WB	18.1	18.6	17.8	16.5	15.0	13.2	12.2	12.2	12.9	13.6	15.1	17.3	(29)
(30)			MCDWB	19.6	20.1	18.8	17.2	15.5	13.7	12.7	12.8	13.5	14.7	16.6	18.7	(30)
(31)		5%	WB	17.4	17.9	17.2	15.8	14.3	12.6	11.6	11.6	12.4	13.1	14.4	16.5	(31)
(32)			MCDWB	18.7	19.2	18.1	16.5	14.9	13.2	12.1	12.2	13.1	14.1	15.6	17.8	(32)
(33)		10%	WB	16.7	17.2	16.6	15.1	13.6	11.8	10.9	10.8	11.8	12.5	13.7	15.7	(33)
(34)			MCDWB	17.9	18.4	17.6	15.9	14.3	12.5	11.6	11.5	12.5	13.5	14.9	17.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.7	5.4	4.9	4.4	4.4	3.8	4.1	4.1	4.5	5.0	5.3	5.4	(35)
(36)			MCDBR	6.8	6.4	5.7	5.0	5.0	4.2	4.6	4.7	5.1	6.1	6.5	6.3	(36)
(37)			MCWBR	4.6	4.2	4.0	4.1	4.5	3.9	4.1	3.9	4.2	4.6	4.5	4.2	(37)
(38)		5% WB	MCDWB	5.5	5.4	4.7	4.5	4.5	3.9	4.3	4.3	4.4	4.9	5.5	5.4	(38)
(39)			MCWBR	4.1	4.1	3.7	4.1	4.2	3.8	4.1	3.9	3.9	4.1	4.2	4.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.362	0.347	0.339	0.332	0.323	0.328	0.321	0.329	0.346	0.348	0.340	0.349	(40)	
(41)		taud	2.499	2.555	2.574	2.575	2.564	2.456	2.516	2.539	2.487	2.489	2.525	2.532	(41)	
(42)		Ebn at Noon	961	948	909	840	769	710	758	828	884	936	977	980	(42)	
(43)		Edh at Noon	111	99	88	75	63	63	65	76	94	105	107	108	(43)	
(44)	All-Sky Solar Radiation	RadAvg	6.25	5.30	3.93	2.53	1.62	1.22	1.47	2.21	3.38	4.71	6.00	6.35	(44)	
(45)		RadStd	0.41	0.36	0.19	0.13	0.09	0.07	0.09	0.15	0.20	0.31	0.40	0.39	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.57	N/A	+0.59	+1.02	+0.43	+0.46	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page